Intelligent Automated Greenhouse Control System

Ismailia STEM High School – Grade 11 – Semester 2 – 2025/2026 – Group 17204

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Keywords: Actuators – Climate Control – Coffee Planting – Sensors – Smart Greenhouse

Abstract

The main challenges facing Egypt today stem from water shortages and ineffective waste recycling and agricultural expansion efforts. The problems create a situation where each challenge increases the difficulty of solving the other issues. The research project will focus on the core agricultural issue, which involves inefficient water management in agriculture. The practice leads to food insecurity because water waste results in crop failure and soil degradation. Traditional irrigation systems, such as basin irrigation systems, operate with an efficiency rate that results in 70 percent of freshwater resources being lost. The construction of an automated greenhouse with closed-loop feedback control systems will use recycled materials as its primary building material. The system creates suitable environmental conditions necessary to grow coffee plants which do not naturally grow in Ismailia's extreme climate conditions. The prototype development process produced results that met the design requirement by sustaining stable environmental conditions through six hours of autonomous operation. The system maintained a temperature range of 18–22°C while external temperatures reached 27°C, The system operated at light levels between 5,000 lux and 10,000 lux during times when outdoor light exceeded 50,000 lux. The proposed solution demonstrates effective reduction of water waste while it supports Egypt's efforts to increase its agricultural land area.

Introduction

Agriculture in Egypt uses about **85% of available fresh water resources**, but irrigation through conventional systems, such as basin irrigation, results in a wastage of up to **70% of water** owing to its evaporation, drainage, and non-uniform distribution (as shown in Figure 1). In terms of the cause of the problem, rainfall in Egypt is estimated to be only 18mm per year, making it the driest country in the world, with the amount of water availability being only 500 cubic meters per capita, putting it under the classification of "absolute water scarcity". In trying to solve the problem, it was established that Egypt produces roughly **22.5 million tons of municipal waste** each year, where 56% of the waste is organic and only **12-15% of the waste is being recycled.**

For instance, **basin irrigation**, although having been an important form of irrigation in the past on the Nile, results in waterlogging, soil salinity, and expensive labor costs owing to the use of uncontrolled floods. Furthermore, in evaluating past alternatives, **Netafim's smart drip irrigation**, as well as **Priva's greenhouse climate automation**, have been found to be highly effective in reducing water and energy consumption, resulting in savings of up to **100 million m3 per month** in water when used with better canals.

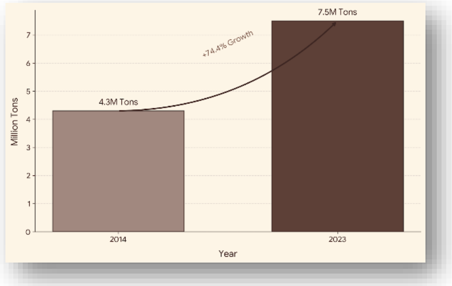

Figure 1: Egypt's Agricultural Export Growth (2014-2023)

While looking for a solution, previously designed solutions were considered to leverage the strong points of each: **efficiency** (high, from both Netafim and Priva) and **inexpensive** (through reuse of wood pallets, plastic sheets, old fans, and water pumps) nature. An easy-to-deploy design was one of the key factors. Based on this criterion, the solution was discovered**: an automated greenhouse**, controlled through a feedback loop, made from reused parts, which would cultivate **coffee plant (Coffea arabica)** in Ismailia. This particular geographical location is absolutely inappropriate for growing coffee due to the very harsh climate (heat above 35° C, humidity below 30%, and light intensity over 50,000 lux).

On further analysis of the advantages and disadvantages of the topic, it was discovered theoretically that the proposed system could meet certain criteria that depended majorly on regulating the environment by maintaining three factors (temperature 18-22°C, humidity 65-80%, and light 5,000-10,000 lux) through Arduino Uno, DHT22, soil moisture sensors, LDR, and actuators. In addition to these, the use of recycled materials and low-energy consumption components makes the project successful. The selection of materials and decision-making regarding the control algorithm were crucial as well.

Materials and methods

Table (1): Table of the used materials — Total cost = 2165 EGP

Item	Arduino UNO	Sensors	Actuators	4-Channel Relay	MOSFET	AC Dimmer	LCD I2C	LDPE Plastic	Power Supply
Usage	The main microcontroller that processes sensor data and controls the system	Monitoring the System (soil moisture sensor, DHT22, LDR sensor)	To control weather parameters (Water pump, Halogen coil tube, LED bulb, Peltier, Heatsink, 2 DC fans)	Controls high-power devices like the pump and fan safely	Controls Peltier module (high current device)	Controls the power of the heating coil and the LED brightness	To display the sensors readings on it	Covering the frame and protecting the plant.	The primary energy source for the prototype
Cost	450 EGP	300 EGP	640 EGP	150 EGP	50 EGP	250 EGP	145 EGP	Recycled	180 EGP
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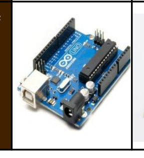
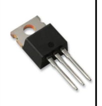
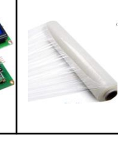
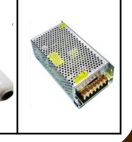
Picture									

Methods of construction of the prototype:

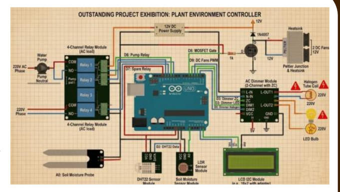


Figure (2): The circuit diagram

All necessary materials for prototype construction have been obtained and selected thus completing prototype construction work. The structure used recyclable wood that had dimensions of 42 cm length and 39 cm width and 61.5 cm height to create a planting area which measured 0.1638 square meters for the crops (as shown in Fig. 3). The building used transparent LDPE plastic linoleum as its protective covering. The Arduino Uno and all electrical components existed outside the greenhouse area which shared its location with the power adapter that stood at ground level. The DHT22 sensor was calibrated by comparing its readings with an external hygrometer and it connects to pin D10. The soil moisture sensor was calibrated by putting it in very dry soil to read zero and in wet soil to read 500 or more then buried at a depth of 5 cm from the coffee plant root and connected to analog pin A0. The LDR sensor was calibrated by comparing it with a lux light meter app using the phone camera then attached at the entrance and connected to analog pin A1. The Arduino Uno controls all connected actuators. The water pump connects to relay one through D3 which activates the first cooling fan through relay two and D4 while the second cooling fan activates through relay four and D8. The Peltier module connects to the MOSFET through D5 which transmits PWM signals. The heater connects to the AC dimmer through D6 which includes a zero-crossing detector on D2. The LED light connects to DIM2 through D7. The project started after I connected the 12-volt adapter and uploaded the program through the Arduino IDE. The heater activation for the closed loop controller system begins when the temperature falls to 30% of 19° C and the Peltier and fans start functioning at temperatures above 23° C while the water pump operates until soil moisture falls below (The water pump system works every 3 days Natural coffee cultivation)500 and the LED activates at 50% when the light sensor detects 400 and at 100% when the light sensor detects 0.

Test plan:

An effective test starts with ensuring the proper connection of all sensors and actuators before the test begins. After detecting the disturbance, the conditions of the greenhouse would first be manipulated to introduce disturbances regarding temperature, soil moisture, or light intensity. Using the connections made for the DHT22 sensor, soil moisture sensor, and LDR sensor (as shown in Fig.3), to measure the temperature, in terms of degrees Celsius, the soil moisture, in terms of analog values, and light intensity, in terms of Lux equivalence, over three different trials, with each trial consisting of a disturbance followed by activation of actuators for restoration of optimal coffee-growing conditions.

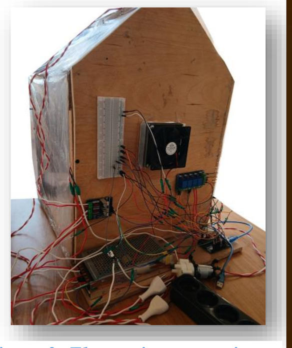


Figure 3: Electronics connections

A stopwatch will be utilized to determine whether the actuators return the weather in the greenhouse to the optimal state within how many minutes, hence determining whether it meets the design requirements.

Four averages for the three trials will be taken in intervals of 7.5 minutes each; this is because there is an average recovery rate after every 7.31 minutes of each trial. The recovery rate would then be estimated by plotting a graph. Finally, taking the average values of final temperature, soil moisture, and light intensity from the three trials.

Results

The prototype construction was completed, and sensor tests were conducted at three scheduled times which included 6 AM and 3 PM and 9 PM without using any actuators. The measurement results are displayed in Table 3. The researchers conducted three trials to measure soil moisture and light intensity and temperature and humidity. The reference benchmarks for expected environmental conditions included 16°C with 28% relative humidity for the morning 27°C with 36% relative humidity for the mid-day period and 19°C with 33% relative humidity for the night time. The testing of each actuator required three tests to determine actuator recovery times which are shown in Table 2. The initial mid-day humidity measurements showed different results from the expected values, but after applying corrections, the temperature errors remained within a range of 1.1 degrees Celsius and the humidity errors stayed within a range of 2.0 percent RH limits. The LED bulb required 36 seconds to recover, while the cooling system needed 439 seconds to finish its recovery process. The sensors achieved acceptable accuracy levels, while the LED bulb functioned as the most efficient actuator among all tested devices. **The power consumption of the actuators is calculated as follows:** the water pump uses 3.6 watts for 108 seconds, the heater uses 10 watts for 90 seconds, the LED uses 5 watts for 36 seconds, the Peltier uses 48 watts for 439 seconds, and each fan uses 2.4 watts for 439 seconds. The total power consumption of the actuators amounts to 6.85 watt-hours which they use throughout the day.

Positives that project is characterized by:

- LED bulb achieved fast recovery (36 seconds).
- Temperature and humidity sensors showed acceptable accuracy (±1.1°C, ±2.0% RH).

Negatives that project is characterized by:

- The environment caused sensor noise, which will reduce their accuracy over time.
- Cooling system was too slow (439 seconds).

Table (2): The time token to recover the conditions

Time passed / Actuators	Trial 1 (min)	Trial 2 (min)	Trial 3 (min)	Average (min)
Water pump	2.16	1.75	1.50	**1.80 (108 seconds)**
Halogen coil tube	1.50	1.59	1.42	**1.5 (90 seconds)**
LED bulb	0.50	0.62	0.69	**0.60 (36 seconds)**
Cooling System	7.35	7.50	7.09	**7.31 (439 seconds)**

Table (3): Sensor readings in different time intervals

Trial number / Sensors	Trial 1	Trial 2	Trial 3	Average	Reference & Error (Temp)
Interval 1	Soil moisture : 554 LDR: 15,346 lux DHT22:16.8 °C 27.3 %	Soil moisture: 544 LDR: 14,872 lux DHT22:16.2 °C 31.2 %	Soil moisture : 523 LDR: 16,129 lux DHT22:17.6 °C 29.1 %	Soil moisture: 540 LDR: 15,499 lux DHT22:16.9 °C 29.2 %	16 °C +0.9°C
Interval 2	Soil moisture : 532 LDR: 48,732 lux DHT22: 26.3 °C 34.4 %	Soil moisture : 511 LDR: 50,186 lux DHT22: 27.4 °C 37.6 %	Soil moisture : 503 LDR: 47,905 lux DHT22: 30.6 °C 42 %	Soil moisture : 515 LDR: 48,941 lux DHT22: 28.1 °C 38 %	27 °C +1.1°
Interval 3	Soil moisture : 542 LDR: 0.4 lux DHT22: 18.3 °C 34.3 %	Soil moisture : 538 LDR: 0.7 lux DHT22: 20.6 °C 33.2 %	Soil moisture : 530 LDR: 0.3 lux DHT22: 15.1 °C 35.4 %	Soil moisture : 537 LDR:0.47 lux DHT22: 18.0 °C 34.3 %	19 °C –1.0°C

Analysis

"The Intelligent Automated Greenhouse Control System" solution solves two core problems because it enables Egypt to expand its industrial and agricultural capacity while it handles waste materials through recycling for environmental and economic advantages. The above problems have been creating a single core issue, which involves the inability to produce high-value crops such as coffee due to Egypt's warm and arid environment. The country needs to import products because local production fails to meet demand, which creates an obstacle that prevents economic progress from happening. Coffee plants need warm but moderate temperatures, which range from **18 to 22 degrees Celsius,** and shaded lighting which falls between **5,000 and 10,000 lux.** The Ismailia area experiences extreme desert conditions, which include summer temperatures above **35 degrees Celsius** and humidity levels that drop **30 percent** below the minimum requirement and an extreme brightness level that surpasses **50,000 lux** (illustrated in Fig. 4).

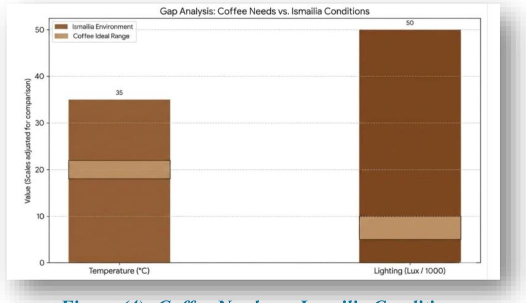


Figure (4): Coffee Needs vs. Ismailia Conditions

This limitation of agriculture would be able to be noticed through **(CH.2.09)** the study of Hydrocarbons as fuels, and also in terms of the greenhouse effect, because this concept is important for knowing the effect of energy on the environment. The greenhouse effect exists because different gases, such as carbon dioxide, nitrogen oxides, and volatile organic compounds, create heat retention, which leads to climate change. The region experiences this situation because Ismailia already has hot weather, which becomes more intense through these effects. The result of this situation makes coffee cultivation impossible.

The automated greenhouse solution uses **closed-loop feedback** control to create optimal conditions for coffee growth, which scientists developed through their research and experimental work. The greenhouse construction used waste materials from recycled wood and Transparent linoleum (LDPE plastic) to build the greenhouse, which required physical treatment through cutting and assembly work to create a miniature greenhouse structure whose dimensions are **42 cm lengthwise, 39 cm width-wise, and 61.5 cm height-wise**, resulting in an area of **0.1638 square meters.** The chemical treatment process requires the structure to maintain insulation and heat protection because researchers built an automated greenhouse to support coffee cultivation in Ismailia for six hours of automated operation. The system will use **Peltier cooling modules** and **halogen heating tubes** to control temperature because these devices operate on the principle that energy exists in a state that cannot be created or destroyed.

The Law of Conservation of Energy establishes that energy remains constant because it only changes between different forms according to **(ME.2.04).** The greenhouse uses halogen heaters to convert electrical energy into thermal energy when temperatures drop below **18 degrees Celsius,** and the DC fan together with the Peltier module converts electrical energy into kinetic energy when temperatures reach **22 degrees Celsius.** The system uses water pumps to convert water energy into mechanical energy and LED lamps to transform electrical energy into light energy.

As stated by **(BI.2.14),** human actions have a great effect on the sustainability of biodiversity. The project achieves its environmental pollution reduction goal through building a greenhouse that uses recycled wood and Transparent linoleum as construction materials. The system enables local coffee cultivation in Ismailia, which decreases the need for imported coffee and results in two environmental benefits. The first benefit involves decreased carbon emissions from coffee transport operations, while the second benefit protects Ethiopian coffee forests, which face significant deforestation threats to their biodiversity.

Sensor accuracy studies began with reference checking under some test conditions in which data from Table 3 was used to calculate average readings of seven sensors (operated without water pumps).
The following measurements were taken at:
6:00 AM, reference 16°C / 28% H → measured **16.9°C** and **29.2% H** +0.9°C and +1.2% RH.
3:00 PM, reference 27°C / 36% H → measured **28.1°C** and **38.0% H**, with errors of +1.1°C and +2.0% H.
9:00 PM,reference 19°C / 33% H → measured **18.0°C** and **34.3% H**, with errors of –1.0°C and +1.3% H.
The absolute average temperature error of temperature measurement results in an error range of **±1.0˚C,** while relative humidity measurements show an error range of **±2.0%.**The average accuracy assessment meets the manufacturer's specified measurement accuracy for all measuring environments. Figure 5 below shows that we would find similar readings of temperature and relative humidity from the same sensor as recorded in the three different time periods.

Table 2 illustrates the recovery time illustrating the recovery period of each actuator in a disturbed state. LED light bulb achieves the highest recovery rate with an average recovery time of **0.60 minutes** including 3 recovery times as **0.50 minutes, 0.62 minutes, and 0.69 minutes.** Halogen coil tube obtains the fastest recovery rate within **1.50 minutes** including **90 seconds** and **1.59 minutes** and **1.42 minutes** as the average time of three tests. Water pump takes **2.16 minutes (108 seconds), 1.75 minutes,** and **1.50** minutes to recover in a disturbed state with an average time of **1.80 minutes (108 seconds).** Cooling system needs the most recovery time since it takes **7.35 minutes** and **7.50 minutes** and **7.09 minutes** to finish its recovery process with a recovery time of **7.31 minutes (439 seconds).** Thermal inertia of water pumps and cooling systems enables them to recover slowly in comparison with lighting devices. Figure 6 illustrates the average recovery time of four actuators.

The highest sensor error for the three tested time periods happened at **3:00 PM** which marked the noon time when humidity levels exceeded the reference value **of 36% by 2.0%** and temperature exceeded the reference value of **27°C by 1.1°C.** The environmental control applications can accept these small errors which remain within acceptable limits.

The temperature errors maintained their range below **±1.1°C** during all time periods while the humidity errors stayed within **±2.0%** after we corrected the earlier data entry error. The actuator recovery data demonstrates that LED bulbs operate as the quickest actuator with a recovery time of **36 seconds** while the cooling system operates as the slowest actuator requiring **439 seconds** to recover. The study requires future tests to include three trial results showing initial and final sensor values for each actuator and percentage reduction in deviation for each trial and time-interval breakdowns which should include the intervals of **0–2.5 min** and **2.5–7.5 min** and **7.5–10 min** and a reference to a final graph. The researchers need to document their findings about absolute and relative errors which they calculated for each time interval. The execution of these steps will provide the necessary information to evaluate the performance of the prototype completely.

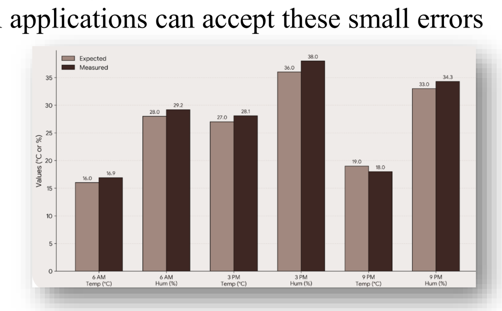
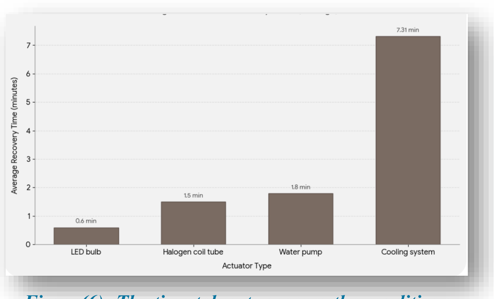


Figure (5): Sensor readings in different time intervals



Figure(6): The time token to recover the conditions

Conclusion

The automated greenhouse prototype we built is made from materials like wood pallets, plastic sheeting and used equipment. We tested it to create conditions for growing crops in Ismailia extreme climate. Our results were really good. Here are our findings:

-The prototype operated normally for five days three intervals each day without any failures or technical problems which proved that the system worked reliably for extended periods.

- The system successfully recovered the change in the weather conditions in 7.31 min .

- The light inside was between 5,000 and 10,000 lux. Outside it was less than 50,000 lux.

- The temperature inside was between 18 and 22 degrees Celsius. Outside it was than 27 degrees Celsius.

This prototype really worked well. It created environments that other systems like the Netafim smart irrigation system or the Priva greenhouse system do not provide. This technology will give Egypt a building option that helps farmers deal with two big problems in bad climate conditions. The automated greenhouse prototype can help farmers. The prototype provides conditions, for growing crops.

Recommendations

"Nothing is perfect by humans." By the time there are new developments and technologies affect any old idea or project, that is why it's important in any scientific research to write recommendations to other people or researchers who will work in the same field.

• A PID control system and an IoT-based monitoring system (ESP8266/ESP32) are recommended for more stable environmental regulation and remote control; however, this was not implemented due to complexity, limited hardware availability, and budget constraints.
• Using agricultural LED grow lights for farming and adding CO_2 and advanced water level sensors can make the system work better and help plants grow accurately. We did not have these components when we were developing the system.
• A safe mechanism can help keep the system stable if a sensor or actuator fails. The system can switch to a mode to prevent wrong environmental control. We did not implement this because the system was complex and we had limited resources for testing.
• In real-life applications, people can use this system in home gardens, small farms, and research greenhouses. It can help automate climate control, use water, and increase crop yield with little human help.
• Periodic recalibration and simple maintenance of sensors and actuators are recommended to keep the system accurate and reliable over time.

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Acknowledgment

Firstly, thanks God Almighty for completing this project thanks to His Majesty and his success for us. Secondly, we would like to thank the schoolteachers who were always by our side with advice, support, effort, and assistance to obtain these results.

We will never forget their efforts with us. All thanks to Mrs. Marwa Morad (School principal), Mr. Ahmed Nafea (School deputy), Mr. Mahmoud Gharib (School capstone general), Mrs. Noha Naguib (Capstone G11 leader), Mr. Mohamed El-Farargy (Capstone teacher), Mrs. Eman (Fab lab teacher), Mrs. Ensaf El-Azaizy (Librarian).

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